

Cognitive Security: Adversarial Robustness and Type Safety

Quantum Cryptographic Communications and Semantic Security

The quantum topological framework of ?? opens a natural connection to quantum cryptographic communications, with implications for both the security and the semantic integrity of case-structured information transfer.

Quantum Key Distribution for Relational Semantics

Quantum key distribution (QKD) protocols provide information-theoretic security guarantees that classical cryptography cannot achieve [pirandola2020qkd]. When case-marked relational structures are communicated between agents—whether human or artificial—QKD ensures that the *relational semantics* of a message (who does what to whom) cannot be intercepted or altered without detection. This is particularly relevant for high-stakes domains where case assignment carries legal or medical significance: a tampered case frame could change an agent's interpretation of responsibility, causation, or obligation.

Cognitive Security: Protecting Relational Reasoning

The cognitive integration of ?? raises a complementary concern: *cognitive security*—ensuring that an active inference agent's generative model of relational structure is not adversarially corrupted. In the predictive processing framework, an agent's case-assignment system is a generative model that minimizes variational free energy. Adversarial manipulation of this system could:

- ▶ **Inject false case frames:** Leading an agent to misidentify the agent, patient, or instrument of an action—a form of semantic adversarial attack
- ▶ **Exploit precision weighting:** Artificially inflating the precision of misleading sensory evidence, causing the agent to update its case assignments toward adversarially chosen interpretations
- ▶ **Corrupt topos-theoretic transfer:** If an agent uses Morita equivalence to transfer case-theoretic results between frameworks, corrupting one framework could propagate errors across all equivalent formulations

Prompt Injection as Adversarial Case-Frame Manipulation

The case-theoretic framework offers a novel structural analysis of *prompt injection attacks*—the predominant vulnerability in contemporary LLM-based agent systems [arXiv:2501.01987]. From the case-theoretic perspective, a prompt injection attack is fundamentally an attempt to *re-assign the case roles* of participants in the human-AI interaction:

In a well-formed agent interaction, the relational structure is:

- ▶ The **user** occupies NOM (Agent)—the initiator of requests
- ▶ The **system prompt** occupies LOC (Context)—the boundary conditions governing behavior
- ▶ The **AI model** occupies INS (Instrument)—the means of executing the user's intentions
- ▶ The **tool/API** occupies ACC (Patient)—the target of directed operations
- ▶ The **output** occupies DAT (Recipient)—the endpoint of information transfer

A prompt injection attack succeeds by *covertly re-assigning* these

Case-Theoretic Firewalls for Multi-Agent Communication

The case-role analysis of prompt injection suggests a principled defense: **categorical type-checking at agent communication boundaries**. Just as a type-safe programming language prevents category errors at compile time, a case-theoretic firewall would enforce relational type constraints on every message exchange:

1. **Case-role immutability:** Once a participant is assigned a case role (NOM, ACC, INS, LOC) at the protocol level, no subsequent message content can alter that assignment. This is enforced by requiring that the case-type of each wire in the interaction diagram is fixed at connection time—analogueous to the type discipline in pregroup grammar, where each word receives its type before composition begins.
2. **Relational integrity constraints:** Every message must type-check against the interaction diagram's expected morphism types. A response from an ACC-typed data source that contains NOM-typed command structures would be rejected as a type error, preventing the case-role promotion

Epistemic Case Security in Multi-Agent Systems

The interaction between case theory and cognitive security acquires particular urgency in multi-agent AI ecosystems where agents must reason about each other's beliefs, intentions, and relational roles. We propose *epistemic case security* as a framework for protecting the relational reasoning of AI agents operating in adversarial environments.

In a multi-agent system governed by case categories, each agent maintains a generative model (in the active inference sense) of the case-frame structure of its interactions. This model determines who is acting (NOM), who is acted upon (ACC), what tools are being used (INS), and what contextual constraints apply (LOC). An adversary targeting the epistemic level of this system does not merely inject false data—it attempts to *corrupt the agent's generative model of relational structure itself*.

- **Belief injection:** Causing an agent to adopt a false case-frame interpretation of observed interactions—believing that agent A (NOM) is acting on agent B (ACC), when in